

Sushant S. Mahajan

Solar Astrophysicist • Data Scientist • AI for Science
solarmagnetism.org ORCID: 0000-0003-1753-8002

SUMMARY

Research scientist and computational physicist building data-driven systems for solar physics, scientific inference, and time-series analysis. Experience spans observational data products, large-scale signal extraction, forecasting workflows, reproducible software, and cross-functional research communication.

EXPERIENCE

Research Scientist, W. W. Hansen Experimental Physics Laboratory, Stanford University 2024–present

- Lead research on solar dynamics and large-scale flows from HMI/SDO observations.
- Produce new metadata and analysis workflows for solar magnetogram and velocity-field data.
- Contribute to an agentic AI-powered JSOC data-commons effort for public query and visualization.

Postdoctoral Fellow, W. W. Hansen Experimental Physics Laboratory, Stanford University 2022–2024

- Published first-author work on differential rotation, torsional oscillation, and meridional flow.
- Helped deliver the Stanford Solar Observatories Group data-sharing effort recognized in 2026.
- Collaborated across research, software, and data management teams on mission-facing analysis.

Solar Physics Postdoctoral Fellow, Institute for Astronomy, University of Hawai'i 2019–2022

- Built analysis pipelines for active-region inflows, solar-cycle trends, and helioseismic flow inference.
- Contributed to forecasting-oriented work on rare events and solar flare data.

SELECTED IMPACT

- NASA Silver Group Achievement Award for JSOC flood recovery efforts 2025
- Stanford University Libraries Data Sharing Prize, Stanford Solar Observatories Group 2026
- Editor's Choice, *Solar Physics*, for *The Sun's Large-Scale Flows I* 2024
- Prepared and published the SWAN-SF benchmark dataset used in solar-flare AI work 2020
- Experience spanning Python, scientific computing, rare-event prediction, and large observational datasets

SELECTED PUBLICATIONS

Mahajan, S. S., Upton, L. A., Antia, H. M., Basu, S., DeRosa, M. L., Hess Webber, S. A., et al.
The Sun's Large-Scale Flows I: Measurements of Differential Rotation & Torsional Oscillation. *Solar Physics* 299, 38 2024

Mahajan, S. S., Sun, X., and Zhao, J.
Removal of Active Region Inflows Reveals a Weak Solar Cycle Scale Trend in the Near-Surface Meridional Flow. *Astrophysical Journal* 950, 63 2023

Dash, S., DeRosa, M. L., Dikpati, M., Sun, X., **Mahajan, S. S.**, Liu, Y., and Hoeksema, J. T.
Ensemble Kalman Filter Data Assimilation into the Surface Flux Transport Model to Infer Surface Flows. *Astrophysical Journal* 2024

SKILLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, scientific computing, time-series analysis, machine learning workflows, data analysis, visualization, and technical writing.

EDUCATION

<i>Ph.D.</i> in Astronomy, Georgia State University	2019
<i>M.S.</i> in Physics, Georgia State University	2017
<i>M.Tech.</i> in Engineering Physics, IIT (BHU)	2014